HCL P_311 INDUSTRIAL MOTOR FAULT DIAGNOSIS MANUAL
Document ID: MAN-IND-2026-V3 | Classification: Industrial Maintenance

SECTION 1: EQUIPMENT OVERVIEW & THERMAL GUIDELINES
1.1 Scope and Machinery Classes:
This technical standard applies to Class II and Class III medium induction motors (10kW - 45kW), specifically Centrifugal Pump Motors (MOTOR-PUMP-01), Heavy Belt Conveyors (MOTOR-CONV-02), and Twin-Screw Compressors (MOTOR-COMP-03).

1.2 Thermal Thresholds & Overheating Fault Mode:
- Continuous Safe Operating Zone: 30°C to 75°C casing temperature.
- Advisory Warning Zone: 75°C to 89°C. Causes: cooling fan shroud clogged with airborne lint/dust, high ambient temperature exceeding 40°C, or grease viscosity breakdown.
- Critical Shutdown Limit: 90°C and above. Operating above 90°C risks rapid thermal degradation of Class F winding insulation (NEMA Standard MG-1) leading to turn-to-turn electrical shorts.
Recommended Action: Immediately inspect cooling fan cowl, verify ventilation airflow, inspect grease relief port for dried grease blockages, and check stator RTD sensors for phase imbalance.

SECTION 2: VIBRATION ANALYSIS & ISO 10816-3 CRITERIA

2.1 ISO 10816-3 Severity Zones for Rigid/Flexible Mounted Motors:
- Zone A / B (Good / Satisfactory): 0.0 - 2.8 mm/s RMS. Smooth mechanical operation.
- Zone C (Warning): 2.8 - 4.5 mm/s RMS. Unrestricted long-term operation not permitted.
- Zone D (Critical): Above 7.1 mm/s RMS. Machine damage imminent. Trip equipment.

2.2 Diagnostic Indicators for Elevated Vibration:
- 1X RPM Dominant Peak: Mechanical unbalance of rotor or cooling fan impeller.
- 2X RPM Peak with High Axial Vibration: Angular or parallel shaft misalignment at coupling.
- High-Frequency (>1000 Hz) Non-Synchronous Peaks: Rolling element bearing wear.
  Early signs include micro-spalling on bearing inner raceway (BPFI) or ball elements (BSF).
Recommended Action: Lockout-tagout (LOTO) motor, perform dial indicator or laser shaft alignment, check foundation bolt torque (tighten to 85 Nm), and replenish Polyrex EM synthetic bearing grease. Replace bearing assembly (SKF 6308 / 6310 deep-groove) if vibration persists above 7.1 mm/s.

SECTION 3: ELECTRICAL OVERLOAD & POWER QUALITY ANOMALIES

3.1 Overcurrent Diagnostics:
- Rated full-load amperes (FLA): MOTOR-PUMP-01 (28A), MOTOR-CONV-02 (21A), MOTOR-COMP-03 (41A).
- Warning Threshold: Amperage > 115% FLA for > 15 seconds.
- Critical Threshold: Amperage > 135% FLA. Risk of immediate thermal breaker trip or stator damage.
Root Causes: Mechanical jamming on conveyor belt, discharge valve shutoff (deadheading) on pump, viscous slurry buildup, or single-phasing condition.

3.2 Supply Voltage Sags & Bus Drops:
- Nominal supply voltage: 230V line-to-neutral (400V line-to-line 3-phase AC).
- Voltage Sag Warning: Supply voltage drops below 207V (-10%).
- Voltage Sag Critical: Supply voltage drops below 195V (-15%).
Under voltage sag, induction motors draw proportionately higher current to sustain shaft torque (Torque is proportional to V^2). This triggers sudden stator overheating without mechanical fault.
Recommended Action: Check MCC contactor contacts for pitting, verify transformer tap settings, measure individual phase currents with clamp meter, and confirm soft-starter / VFD bus voltage.

SECTION 4: ACOUSTIC NOISE & CENTRIFUGAL PUMP CAVITATION

4.1 Acoustic Decibel Limits:
- Normal baseline: 65 dB - 76 dB.
- Warning Limit: 85 dB (OSHA 8-hour continuous threshold).
- Critical Limit: 95 dB. High risk of mechanical destruction or hearing hazard.

4.2 Centrifugal Pump Cavitation Phenomena (MOTOR-PUMP-01):
Cavitation occurs when suction pressure drops below liquid vapor pressure, forming vapor bubbles that collapse violently against the pump impeller and casing.
Symptoms: Loud rattling / gravel noise (>88 dB), erratic discharge pressure drop below 3.0 bar, and localized vibration spikes.
Remedy: Clear suction strainer, verify suction valve is 100% open, check net positive suction head (NPSH), and purge entrapped air pockets.

4.3 Compound Bearing Failure (Vibration > 5.5 mm/s AND Temperature > 80°C):
This combination indicates total breakdown of boundary lubrication film. Metal-to-metal contact is actively welding microscopic asperities, leading to rapid bearing cage disintegration and shaft seizure.
Emergency Procedure: Immediate emergency shutdown. Do not attempt restart without bearing replacement.